Purna Mummani

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Summary

Software Engineer with 3 years building scalable C#/.NET backend services and developer productivity solutions on Azure, specializing in CI/CD reliability, workflow automation, and centralized observability. Proficient in Python, Java, and JavaScript deploying production services on AKS and Docker via GitHub Actions CI/CD, with hands-on experience leveraging AI-assisted tools for engineering efficiency.

Technical Skills

Languages: Python, Java, C#, C++, JavaScript, TypeScript, SQL
Frameworks: .NET, ASP.NET Core, React, Node.js, Spring Boot, PyTorch, FastAPI, TensorFlow
AI & ML: Azure ML, Azure OpenAI Service, Hugging Face Transformers, LangChain, RAG, NLP, LLMs, GitHub Copilot
Cloud & DevOps: Azure (AKS, DevOps, App Services, Monitor, Functions), Docker, Kubernetes, GitHub Actions
Tools & DBs: Git, Visual Studio, VSCode, Kafka, Maven, Jupyter, JIRA, Postman, SQL Server, PostgreSQL, Redis, Cosmos DB
Competencies: REST APIs, Microservices, Workflow Orchestration, CI/CD, Azure Pipelines, OOP, Agile, SDLC, Data Structures

Experience

IBM Jun 2025 – Present
Software Engineer *USA*

- Designed developer productivity tooling and workflow orchestration in C#/.NET, reducing manual toil by 40% via self-service automation.
- Strengthened CI/CD pipelines via Azure DevOps and GitHub Actions with automated quality gates, cutting deployment failures by 35%.
- Built AI-assisted diagnostics using Azure OpenAI Service and GitHub Copilot, reducing incident resolution time by 42%.
- Diagnosed complex cross-platform issues across Windows and Linux, enabling end-to-end observability via Azure Monitor.
- Created reusable CI templates with testing automation and dependency drift detection, improving CI signal-to-noise and reducing rollback rate by 27%.

University of North Carolina at Charlotte May 2024 – Apr 2025
Research Assistant — AI/ML for Satellite Image Analysis *Charlotte, NC*

- Deployed Vision Transformer inference on Azure via Docker/Kubernetes, achieving 91% accuracy on 8K+ samples.
- Built FastAPI backends with Azure OpenAI and LangChain RAG, improving retrieval quality by 35% via GitHub Actions CI/CD.
- Optimized mixed-precision GPU training on Azure, cutting model training time by 25% via automated dependency management.

Accenture Jan 2022 – Dec 2023
Software Engineer *Hyderabad, India*

- Architected scalable backend services in C#/.NET and Java on Azure App Services, supporting 1M+ transactions/day at 99.9% uptime.
- Automated CI/CD via Azure DevOps and GitHub Actions with Unit/Mockito and safe rollback strategies, achieving 85% test coverage.
- Contributed engineering documentation and debugging workflows across Windows and Linux, reducing incident response time by 25% via Azure Monitor.

Projects

AI-Powered Video Highlights & Summarization Platform | *Python, React, Azure Functions, GitHub Actions, Azure Monitor*

- Engineered an AI-driven video summarization service in C#/.NET on Azure, processing 8K+ videos via event-driven Azure Functions with Python ML inference, reducing manual review by 55%.
- Built React/TypeScript dashboard with Cosmos DB sub-200ms retrieval and centralized observability via Azure Monitor with safe deployment through Azure DevOps CI/CD.

Intelligent Health Claims Management System | *C#, .NET, React.js, Azure DevOps, , Azure App Service, SQL Server*

- Built a full-stack .NET + React platform on Azure with workflow orchestration via Azure Functions, processing 50K+ claims/day and cutting manual review by 60%.
- Automated CI/CD via Azure DevOps with quality gates and safe deployment strategies, reducing release cycles by 45%.
- Designed event-driven claim validation workflows, cutting manual review by 60% and achieving 85% test coverage.

AI Document Chat Assistant with RAG | *PyTorch, Azure ML, FastAPI, Docker, AKS, GitHub Actions*

- Trained transformer classification models on Azure ML with automated evaluation checks, achieving 93% F1 score on 75K+ documents.
- Built LangChain RAG pipeline with Azure OpenAI enabling conversational Q&A with 85% answer relevance via CI-driven quality gates.
- Containerized inference services on AKS with cross-platform dependency management, reducing latency by 50% via GitHub Actions CI/CD.

Education

University of North Carolina at Charlotte *GPA: 3.8/4* Jan 2024 – Dec 2025
Master of Science in Computer Science *Charlotte, NC*

Certifications: *Generative AI with Large Language Models (DeepLearning.AI) | Machine Learning (Stanford University — Coursera) | Java Full Stack Developer (Coursera) | Data Structures and Algorithms (Coursera) | Prompt Engineering (LinkedIn)*